The names of the groups can then be found by looking up the CN (Common Name) attribute on the results.

> **MINI-PROJECT** Adapt the `UserController` to authenticate users to an LDAP server of your choosing. Appendix A details how to set up an example LDAP server, and the code repository that accompanies this book has an example if you get stuck.

Listing 8.3 Looking up LDAP groups for a user

```
import javax.naming.*;
import javax.naming.directory.*;
import java.util.*;

private List<String> lookupGroups(String username)
        throws NamingException {
    var props = new Properties();
    props.put(Context.INITIAL_CONTEXT_FACTORY,    #A
             "com.sun.jndi.ldap.LdapCtxFactory");    #A
    props.put(Context.PROVIDER_URL, ldapUrl);    #A
    props.put(Context.SECURITY_AUTHENTICATION, "simple");
#A
    props.put(Context.SECURITY_PRINCIPAL, connUser);    #A
    props.put(Context.SECURITY_CREDENTIALS, connPassword);
#A

    var directory = new InitialDirContext(props);    #A

    var searchControls = new SearchControls();
    searchControls.setSearchScope(
            SearchControls.SUBTREE_SCOPE);
    searchControls.setReturningAttributes(
            new String[]{"cn"});
```